Days 1-5: Build and Configure a Small Office Network Deliverable

1.Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.

Aim: To configure static IP addresses on five laptops or desktops in the same subnet and verify connectivity using the ping command.

1. IP Address Configuration Table

Use the following example configuration for a home or lab network.

Device	IP Address	Subnet Mask
PC-1	192.168.1.11	255.255.255.0
PC-2	192.168.1.12	255.255.255.0
PC-3	192.168.1.13	255.255.255.0
PC-4	192.168.1.14	255.255.255.0
PC-5	192.168.1.15	255.255.255.0

All five devices belong to the same subnet, 192.168.1.0/24. Each device has a unique IP address.

2. Step-by-Step Procedure

1. Connect all five devices to the same network using a switch or Wi-Fi router.
2. Open Control Panel → Network and Sharing Center → Change adapter settings.
3. Right-click the active Ethernet or Wi-Fi adapter and select Properties.
4. Select Internet Protocol Version 4 (TCP/IPv4) and click Properties.
5. Select Use the following IP address.
6. Enter the IP address and subnet mask assigned to that device in the table.
7. Repeat the process for all five devices, ensuring no duplicate IP addresses exist.

3. Verify IP Configuration

Open Command Prompt on each device and execute:

ipconfig

Verify that the IPv4 address and subnet mask match the configuration table.

4. Test Connectivity Using Ping

From PC-1, open Command Prompt and execute these commands:

ping 192.168.1.12

ping 192.168.1.13

ping 192.168.1.14

ping 192.168.1.15

To test connectivity from PC-2 to PC-1, run:

ping 192.168.1.11

Repeat similar tests from the other devices if required.

5. Observation Table

Test	**Expected Result**
PC-1 to PC-2	Reply received
PC-1 to PC-3	Reply received
PC-1 to PC-4	Reply received
PC-1 to PC-5	Reply received

Note: These are expected results. Record the actual output from your devices. If a request times out, check the IP configuration, network connection, and firewall settings.

2.Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

Aim: To configure a DHCP server that automatically assigns IP addresses to at least three devices and verify that each device receives a unique IP address from the correct range.

1. Network Configuration

For this experiment, we use Cisco Packet Tracer with one DHCP server and three PCs connected through a switch.

Component	Configuration
Network address	192.168.1.0/24
DHCP server IP	192.168.1.2
Subnet mask	255.255.255.0
DHCP pool start IP	192.168.1.100
Maximum clients	10
Default gateway	192.168.1.1

2. Step-by-Step Procedure

1. Open Cisco Packet Tracer.
2. Add one Server, one Switch, and three PCs (PC-1, PC-2, and PC-3).
3. Connect all devices to the switch using Copper Straight-Through cables.
4. Click the server and open Desktop → IP Configuration.
5. Assign the server a static IP address of 192.168.1.2 with subnet mask 255.255.255.0.
6. Open Services → DHCP on the server and turn the DHCP service On.
7. Configure the DHCP pool using these settings:
    - Pool Name: HOME_DHCP
    - Default Gateway: 192.168.1.1
    - DNS Server: 8.8.8.8
    - Start IP Address: 192.168.1.100

    - Subnet Mask: 255.255.255.0
    - Maximum Number of Users: 10
8. Click Add or Save, depending on the Packet Tracer version.
9. For each PC, open Desktop → IP Configuration and select DHCP.
10. Wait for the IP addresses to be assigned automatically.

3. Verify Assigned IP Addresses

On each PC, open Desktop → Command Prompt and run:

ipconfig

Expected IP address allocation:

Device	Expected IP Address	Subnet Mask
PC-1	192.168.1.100	255.255.255.0
PC-2	192.168.1.101	255.255.255.0
PC-3	192.168.1.102	255.255.255.0

Note: These are expected example addresses. The actual allocation order may differ.

4. Test Connectivity

From PC-1, open Command Prompt and execute:

ping 192.168.1.101

ping 192.168.1.102

Successful replies indicate that PC-1 can communicate with the other PCs.

You can also use Services → DHCP on the server to inspect the configured pool and, where supported, the active DHCP bindings.

5. Observation

- All three PCs obtained IP addresses automatically through DHCP.
- Each PC received a unique IP address within the configured range of 192.168.1.100–192.168.1.109.
- The subnet mask was 255.255.255.0 for all devices.
- Ping tests confirmed local network connectivity if replies were received.

**3.Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.

Hint: You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.**

Aim: To configure DNS so that entering a custom hostname, such as musicplayer.local, in a browser opens a specific local web page or device interface.

1. Network Configuration

For this experiment, we use a local web server and map a custom hostname to its IP address.

Component	**Configuration**
Web server IP	192.168.1.10
Client PC IP	192.168.1.11
Custom hostname	musicplayer.local
Web server port	80 (HTTP)
DNS mapping	musicplayer.local → 192.168.1.10

2. Step-by-Step Procedure (Windows Hosts File Method)

This method is suitable for a quick test on a Windows computer without installing a separate DNS server.

1. Connect the client PC and web server to the same local network.
2. Configure the web server with the IP address 192.168.1.10.
3. Start a web server on that device. For example, if Python is installed, open Command Prompt in the folder containing your HTML page and run:

python -m http.server 80

Ensure port 80 is available and permitted by the firewall.

4. On the client PC, open Notepad as Administrator.
5. Open the following file:

C:\Windows\System32\drivers\etc\hosts

6. Add this line at the end of the file:

192.168.1.10 musicplayer.local

7. Save the file.
8. Open Command Prompt and run:
9. ipconfig /flushdns

ping musicplayer.local

10. Open a browser and enter:

http://musicplayer.local

The browser should display the page served by the computer at 192.168.1.10.

3. Configure Access on Other Devices

The hosts-file mapping applies only to the computer where you added it. To access the same hostname from other devices, you can either:

- Add the same hosts-file entry on each client device, or
- Configure a local DNS server, such as dnsmasq, and set connected devices to use it as their DNS server.

For a dnsmasq server on Linux, add this entry to its configuration file, commonly /etc/dnsmasq.conf:

address=/musicplayer.local/192.168.1.10

Restart the service:

sudo systemctl restart dnsmasq

Configure the clients to use the DNS server's IP address, then test the hostname. Ensure the DNS server is reachable and the firewall permits DNS traffic on port 53.

Note: .local is commonly used by multicast DNS (mDNS), so it may conflict with normal unicast DNS resolution on some devices. For a controlled lab, use a suitable internal DNS name, such as musicplayer.home.arpa, if needed.

4. Verification Table

Test	Expected Result
ping musicplayer.local	Resolves to 192.168.1.10
Open http://musicplayer.local	Local web page opens
Access from another configured client	Same page opens
Enter an incorrect hostname	Name resolution fails

Record the actual results after performing the tests.

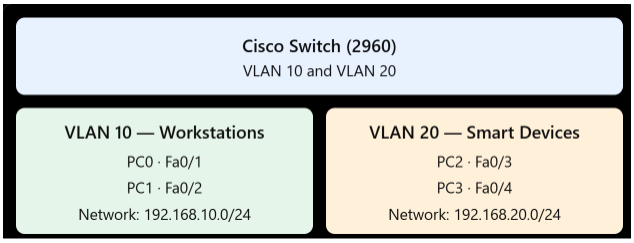
**4.Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.

Hint: Use ping or file sharing to test connectivity between devices in different VLANs.**

Aim: To create two VLANs on a switch, assign at least two devices to each VLAN, and verify that devices within the same VLAN can communicate while devices in different VLANs cannot communicate without inter-VLAN routing.

1. Network Topology

Use Cisco Packet Tracer with one switch and four PCs.



2. IP Address Configuration

Assign static IP addresses to the four PCs through Desktop → IP Configuration.

Device	VLAN	IP Address	Subnet Mask
PC0	10	192.168.10.11	255.255.255.0
PC1	10	192.168.10.12	255.255.255.0

PC2	20	192.168.20.11	255.255.255.0
PC3	20	192.168.20.12	255.255.255.0

3. Configure VLANs on the Switch

Open the switch CLI in Packet Tracer and enter the following commands:

```
enable
configure terminal

vlan 10
name Workstations
exit

vlan 20
name Smart_Devices
exit

interface range fastEthernet 0/1-2
switchport mode access
switchport access vlan 10
exit

interface range fastEthernet 0/3-4
switchport mode access
switchport access vlan 20
exit

end
show vlan brief
```

The show vlan brief command displays the VLANs and their assigned switch ports. Confirm that Fa0/1 and Fa0/2 belong to VLAN 10 and Fa0/3 and Fa0/4 belong to VLAN 20.

4. Test Connectivity Using Ping

Open Desktop → Command Prompt on each PC.

Test A: Same VLAN — PC0 to PC1

ping 192.168.10.12

Expected result: Successful replies because both devices belong to VLAN 10.

Test B: Same VLAN — PC2 to PC3

ping 192.168.20.12

Expected result: Successful replies because both devices belong to VLAN 20.

Test C: Different VLANs — PC0 to PC2

ping 192.168.20.11

Expected result: Ping fails because the devices belong to different VLANs and no inter-VLAN routing has been configured.

5. Observation Table

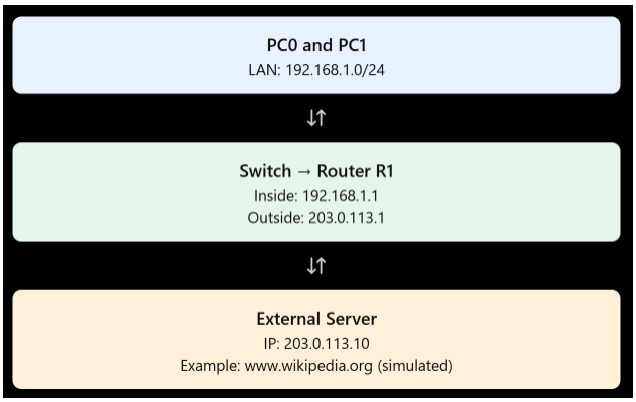
Source	**Destination**	**Expected Result**
PC0 (VLAN 10)	PC1 (VLAN 10)	Successful
PC2 (VLAN 20)	PC3 (VLAN 20)	Successful
PC0 (VLAN 10)	PC2 (VLAN 20)	Failed
PC1 (VLAN 10)	PC3 (VLAN 20)	Failed

5. Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

Aim: To configure Network Address Translation (NAT) on a router so that devices on a Local Area Network (LAN) can access an external website or simulated external server.

1. Network Topology

Use Cisco Packet Tracer with one router, one switch, two PCs, and one external server.



2. IP Address Configuration

Device	Interface	IP Address	Subnet Mask
Router R1	LAN (G0/0)	192.168.1.1	255.255.255.0
Router R1	WAN (G0/1)	203.0.113.1	255.255.255.0
PC0	Ethernet	192.168.1.10	255.255.255.0
PC1	Ethernet	192.168.1.11	255.255.255.0
External Server	Ethernet	203.0.113.10	255.255.255.0

Set the PCs' default gateway to 192.168.1.1 and the external server's gateway to 203.0.113.1.

3. Configure NAT on the Router

Open the router CLI in Packet Tracer. The following example uses Cisco IOS interface names GigabitEthernet0/0 and GigabitEthernet0/1.

enable

configure terminal

interface gigabitEthernet0/0

ip address 192.168.1.1 255.255.255.0

ip nat inside

no shutdown

exit

interface gigabitEthernet0/1

ip address 203.0.113.1 255.255.255.0

ip nat outside

no shutdown

exit

access-list 1 permit 192.168.1.0 0.0.0.255

ip nat inside source list 1 interface gigabitEthernet0/1 overload

end

write memory

Explanation:

- ip nat inside identifies the LAN interface.
- ip nat outside identifies the external-facing interface.
- The access list permits LAN addresses to be translated.
- overload enables PAT (Port Address Translation), allowing multiple LAN devices to share one external IP address using different port numbers.

4. Test Connectivity

On PC0, open Desktop → Command Prompt and run:

ping 192.168.1.1

ping 203.0.113.10

- The first command tests connectivity to the router.
- The second command tests connectivity to the simulated external server through the router.

On the router, verify NAT operation using:

show ip nat translations

show ip nat statistics

show ip interface brief

Generate traffic from PC0 before checking NAT translations so that an entry can be created.

5. Test Website Access

On the external server, enable the HTTP service and configure a web page.

On PC0, open the web browser and enter:

http://203.0.113.10

The simulated external page should open if the interfaces, routes, gateways, and HTTP service are configured correctly.

To test www.wikipedia.org itself, the lab needs a genuine internet connection and working DNS configuration. A Packet Tracer simulated server does not provide access to the real Wikipedia website.

6. Observation Table

Test	**Expected Result**
PC0 pings router LAN IP	Successful
PC0 pings external server	Successful
NAT translation table	Shows translated traffic after a test
Browser opens simulated web page	Successful

Record the actual results from your simulation.